

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

- Appendix on s -wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Superconducting instability

This chapter is devoted to studying the superconducting (SC) phase of the system. Superconductivity is obtained essentially by allowing for Cooper fluctuations, obtaining then a phase much different from normal and AF. The conventional, locally repulsive HM, cannot exhibit a BCS-like superconducting phase, an argument discussed in Sec. ???. The 2D EHM, supplied with non-local attraction, fixes this and makes a SC phase possible, by the means of a slightly more complex Cooper pairing mechanism. Not only, various SC topological classes are possible based on the competition between isotropic and anisotropic pairing. Finally, spin pairing algebra plays a role in the competition defining singlet and triplet pairing channels.

The most interesting effects with respect to conventional HM are the insurgence of a BCS-like superconducting phase, as well as the already discussed bands renormalization and resymmetrization and the extension of the gap function symmetry structure.

1.1 The SC phase and its symmetries

The starting point is, as done for the other phases, the discussion of the symmetries of the SC phase. The BCS-like phase we are dealing with is described by the full $U^c(1)$ SSB. All other symmetries are preserved, but as for the normal and AF phases we allow for

$$C_4 \xrightarrow{\text{SSB}} C_2$$

without breaking inversion symmetry. Therefore, being translational invariance fully preserved, the system density we assume to be uniform across all sites, similarly to the normal phase discussed in Chap. ???.

Recall Tabs. ?? and ??, collecting the RWCs for the AF phase,

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree, Fock and Cooper contractions

We anticipate the role of all RWCs in Tab. 1.1.

Let us highlight now an important point regarding the Cooper class Wick's contractions performed on the general EHM of Eq. (??),

$$\hat{H} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma}}_{\hat{H}_t} + \underbrace{U \sum_i \hat{c}_{i\uparrow}^{\dagger} \hat{c}_{i\downarrow}^{\dagger} \hat{c}_{i\downarrow} \hat{c}_{i\uparrow}}_{\hat{H}_U} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma'}^{\dagger} \hat{c}_{j\sigma'} \hat{c}_{i\sigma}}_{\hat{H}_V}$$

As is discussed in Sec. ??, when applying Wick's theorem the resulting terms break the natural symmetries of the model. The superconducting symmetry we study breaks just the $U^c(1)$ charge conservation. Now, when dealing with superconducting Cooper pairing we need to account also for the spatial structure of the Cooper pair itself. Consider the generic Cooper fluctuation

$$\langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma'}^{\dagger} \rangle \quad \text{with } i, j \text{ NN}$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalization and resymmetrization
		Cooper	Anisotropic pairing in triplet channel

Table 1.1 | Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. ?? and ??.

Spatial structure	Pairing channel	RWCs in Cooper channel
Symmetric wave function	Singlet pairing	Just $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\bar{\sigma}}^\dagger \rangle$
Anti-symmetric wave function	Triplet pairing	Both $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\bar{\sigma}}^\dagger \rangle$ and $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}^\dagger \rangle$

Table 1.2 | Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \rightarrow (-x, -y)$).

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. If we impose a specific spatial symmetry on the hamiltonian the ground state wavefunction will follow naturally, and the pairing channel will be the one providing a total anti-symmetry to the full wavefunction. This gives a selection rule over the relevant pairings: if we work with space symmetric structures – say, s^* -wave or d -wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarized in Tab. 1.2.

1.1.1 Superconductivity in the HM

In Sec. ??, we discuss in detail the (impossibility of a stable) s -wave SC phase in the conventional HM. This is an obvious result, since the two-body interaction thereby present $\hat{H}_U$ is local and repulsive. We nevertheless treat the model within the MFT framework, and here recall the main widely known results, and later use them as groundwork for the EHM extension. Similarly to Sec. ??, let us define the operators

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (1.1)$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \quad (1.2)$$

whose EVs are relevant for the SC phase.

The MFT hamiltonian is given explicitly for the HM in Eq. (??),

$$\begin{aligned} \hat{H}_t + \hat{H}_U \stackrel{\text{MFT}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\downarrow}^\dagger \right] + E_a^{(\text{SC})} \\ - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^\dagger \right] + nU \times \hat{N} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right] \end{aligned}$$

with $E_{\text{H}/U}^{(\text{SC})}$, $E_{\text{C}/U}^{(\text{SC})}$ the contraction energies arising from Hartree and Cooper channels, and $E_a^{(\text{SC})}$ the energy shift resulting from anticommutation rules, given respectively in Eq. (??), (??), and (??).

EV	Meaning	Equation
$\langle \hat{s}_{\mathbf{k}}^x \rangle$	$\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^\dagger \rangle$	(??)
$\langle \hat{s}_{\mathbf{k}}^y \rangle$	$i \langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^\dagger \rangle$	(??)
$\langle \hat{s}_{\mathbf{k}}^z \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} \rangle - 1$	(??)

Table 1.3 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM. The z component refers also to Eq. (??).

In deriving this result we decomposed the interaction of the intermediate passage of Eq. (??) and assumed the Cooper pair to have null total momentum. Moreover, we recognized in the passage of Eq. (??) the presence of a s -wave harmonic transform of the function $w_{\mathbf{q}}$, and thus defined in Eq. (??),

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}} \quad \text{and} \quad \Delta^{(s)} \equiv -U w^{(s)}$$

Nambu-Bogoliubov representation of the MFT hamiltonian is given in Eq. (??) and equivalently in Eq. (??),

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \hat{\Phi}_{\mathbf{k}}^\dagger h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_a^{(\text{SC})} - [E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})}]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}}^* \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$. The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}} \tau^z - \text{Re}\{\Delta_{\mathbf{k}}\} \tau^x - \text{Im}\{\Delta_{\mathbf{k}}\} \tau^y$$

Nambu spinors are, in this case,

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix}$$

For the SC phase in the HM model, in general, we can admit for a complex gap function – although being the gap a constant it can always be reduced to a purely real gap. Diagonalization by the angles defined in Eq. (??) is performed via the composed rotation of Eq. (??),

$$W_{\mathbf{k}} = \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix}$$

The Bogoliubov bands are $\pm E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$. The only crucial difference with the AF Bogoliubov bands is that here, apart from the absence of the spin index – a degeneracy that is solved now by enlarging the computation to the entire BZ –, also the bare bands $\epsilon_{\mathbf{k}}$ are substituted by their shifted version $\xi_{\mathbf{k}}$. This difference is essential: it is this feature that makes the superconducting phase gapped in an insulating sense, which means, opening the gap at the position of $\tilde{\mu}$. Instead for the AF phase the gap was always opened at “midband”, the MBZ contour where $\epsilon_{\mathbf{k}} = 0$. As a final remark, as is proven trivially at the end of Sec. ??, s -wave SC is impossible at MFT level on the pure HM. This was obvious, due to the necessity of an even small attractive interaction to form Cooper pairs. To conclude, in Tab. 1.3 we report the operator-pseudospin map for each direction.

1.1.2 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$: translational invariance is preserved while charge conservation is broken, thus all three channels (Hartree, Fock and Cooper) contribute via RWCs to the

MFT hamiltonian. As for the AF phase, we aim to derive a MFT solution for the SC phase which is an extension of the HM solution of Sec. ??, with an effective redefinition of the various physical quantities and mathematical objects,

$$\xi_{\mathbf{k}} \rightarrow \tilde{\xi}_{\mathbf{k}} \quad E_{\mathbf{k}} \rightarrow \tilde{E}_{\mathbf{k}} \quad \Delta_{\mathbf{k}} \rightarrow \tilde{\Delta}_{\mathbf{k}} \quad \mu \rightarrow \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. ?. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

and as for the AF phase:

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (1.3)$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (1.4)$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (1.5)$$

These equations are identical to the ones for the pure HM, Eqns. (??), (??) and (??). In the tilted frame, we have the analogous of Eq. (??) with the important exception that the chemical potential in the Fermi-Dirac factors is zero,

$$\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f(\tilde{E}_{\mathbf{k}}; \beta, 0) - f(-\tilde{E}_{\mathbf{k}}; \beta, 0) \quad (1.6)$$

This is because all the chemical potential operatorial contributions were absorbed in the bare bands $\tilde{\xi}_{\mathbf{k}}$, thus the pseudospin state is thermally described by an effective null chemical potential.

The (renormalized) thermal factors as in Eq. (??) are

$$\tilde{\text{Th}}_{\mathbf{k}}^{(\pm)}[\beta, \tilde{\mu}] \equiv f(-\tilde{E}_{\mathbf{k}}; \beta, 0) \pm f(\tilde{E}_{\mathbf{k}}; \beta, 0) \quad (1.7)$$

which simply reduce to

$$\begin{aligned} \tilde{\text{Th}}_{\mathbf{k}}^{(+)} &= 1 \\ \tilde{\text{Th}}_{\mathbf{k}}^{(-)} &= \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

For the angles it holds

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \quad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \quad (1.8)$$

From Tabs. ?? and ??, we know all channel contribute. It is already known the effect of the Fock contractions, responsible as for the normal phase (Chap. ??) of the bare bands renormalization. Cooper terms are to be handled more carefully, since can contribute to superconductivity both in the s.s. and in the o.s. spin sectors.

1.2 Non-local Hartree effects: chemical potential renormalization

As always, let us start with a very brief section about the Hartree effects on the MFT decomposition of the hamiltonian. We will not delve in the detailed calculations of the Hartree channel MFT effects, already described in previous chapter as well as in Sec. ?. The only effect here is the chemical potential renormalization.

1.2.1 Hartree channel in the SC phase: analytic derivation

The Hartree fluctuations are density-density couplings in the MFT hamiltonian. Knowing the deal with a uniform-density phase, we can re-use as it is the derivation of Sec. ?. Without repeating the calculations, the $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E_{\text{H/V}}^{(\text{SC})} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E_{\text{H}/U}^{(\text{SC})}$$

gives the chemical potential full renormalization, identical to Eq. (??),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (1.9)$$

1.2.2 Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (??),

$$E_{\text{H}/V}^{(\text{SC})} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (??)

$$E_{\text{H}/U}^{(\text{SC})} = L_x L_y \times n^2 U$$

Now we move to the more interesting Fock sector.

1.3 Fock effects: bare bands renormalization

As done for the other phases, we here include rapidly the bands renormalization effect derived from the Fock sector of the non-local attraction. This discussion is essentially analogous the one done for the normal phase in Sec. ??.

1.3.1 Fock channel in the SC phase: analytic derivation

When it comes to the Fock RWCs, being the SC phase density-uniform we can re-use almost the entirety of the derivation of Sec. ?.?. Setting as in Eqns. (??), (??),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad (1.10)$$

Eq. (??) stands perfectly valid, translated to the SC phase

$$\begin{aligned} & V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right] \\ & \stackrel{\mathcal{F}}{=} \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} u_{\mathbf{k}\sigma} \right] \times V \sum_{\mathbf{k}' \in \text{BZ}} \varphi_{\mathbf{k}'}^{(\gamma)*} \hat{c}_{\mathbf{k}'\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} - E_{\text{F}/V}^{(\text{SC})} \end{aligned} \quad (1.11)$$

being

$$E_{\text{F}/V}^{(\text{SC})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \langle \hat{c}_{\mathbf{k}'\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \rangle \quad (1.12)$$

To get to this point, we employed the assumption

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is obviously well-verified by the BCS ground state of Eq. ?.?. Recognising the symmetry decomposition

$$u_{\sigma}^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$$

due to inversion symmetry the conclusion is identical to Eqns. (??) and (??) (considering spin degeneracy),

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + u^{(s^*)} V (\cos k_x + \cos k_y) + u^{(d)} V (\cos k_x - \cos k_y)$$

In order to estimate the γ -wave components renormalizing the bare bands, we need to compute separately in spin sectors the EVs of $u_{\mathbf{k}\sigma}$

$$\begin{aligned}
u_{\mathbf{k}\uparrow} &= \langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} \rangle \\
&= \left\langle \hat{\Phi}_{\mathbf{k}}^\dagger \frac{\mathbb{I} + \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\
&= \frac{1}{2} \left[\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \hat{\Gamma}_{\mathbf{k}} \rangle + \langle \hat{s}_{\mathbf{k}}^z \rangle \right] \\
&= \frac{1}{2} \left[\tilde{\text{Th}}_{\mathbf{k}}^{(+)}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta] \right] \\
&= \frac{1}{2} \left[1 - \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]
\end{aligned} \tag{1.13}$$

having we used Eq. (1.7) and following. For the spin $\downarrow$ EV,

$$\begin{aligned}
u_{-\mathbf{k}\downarrow} &= \langle \hat{c}_{-\mathbf{k}\downarrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow} \rangle \\
&= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \\
&= 1 - \left\langle \hat{\Phi}_{\mathbf{k}}^\dagger \frac{\mathbb{I} - \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\
&= 1 - \frac{1}{2} \left[\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \hat{\Gamma}_{\mathbf{k}} \rangle - \langle \hat{s}_{\mathbf{k}}^z \rangle \right] \\
&= 1 - \frac{1}{2} \left[\tilde{\text{Th}}_{\mathbf{k}}^{(+)}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta] \right] \\
&= 1 - \frac{1}{2} \left[1 + \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]
\end{aligned} \tag{1.14}$$

From the above results, note first that

$$u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}$$

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. From the above results we have a simple expression for the system density,

$$\begin{aligned}
n &= \frac{N}{2L_x L_y} \\
&= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} u_{\mathbf{k}\sigma} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[1 + \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]
\end{aligned}$$

Moreover,

$$\begin{aligned}
\lim_{\tilde{\theta}_{\mathbf{k}} \rightarrow 0} u_{\mathbf{k}\sigma} &= \frac{1}{2} \left[1 + \tanh\left(\frac{\beta \tilde{\xi}_{\mathbf{k}}}{2}\right) \right] \\
&= \frac{1}{2} \left[\frac{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}} - \frac{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} - e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}} \right] \\
&= f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]$	Fig. ??
$d_{x^2-y^2}$ -wave	$u^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]$	Fig. ??

Table 1.4 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

We can then derive the $u_{\mathbf{k}}$ -related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{n}_{\mathbf{k}\uparrow} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \left[1 - \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]
\end{aligned} \tag{1.15}$$

reported explicitly for the only two inversion symmetric structures in Tab. 1.4.

1.3.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (??) holds identically

$$E_{\text{F}/V}^{(\text{SC})} = L_x L_y \times V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (1.13), (1.14).

1.4 Cooper effects: anisotropic pairing

Anisotropic pairing in superconducting fluctuations is induced by the Cooper RWCs. The two sectors of $\hat{H}_V$ (o.s., s.s.) are Fourier-transformed respectively in Eqns. (??), (??). In Tab. 1.2 we identified the correct RWCs for singlet and triplet pairing channels, noting that the o.s. sector contributes to both while the s.s. to just the triplet channel.

1.4.1 Cooper o.s. channel in the SC phase: analytic derivation

This section deals with Cooper fluctuations induced by the o.s. part of the non-local hamiltonian –referring to the notation of Eq. (??)– somewhat the simplest form of Cooper pairing. This sector contributes both to singlet pairing and triplet pairing. Considering Cooper fluctuations in the singlet channel, we need to break $U^c(1)$ symmetry imposing space inversion symmetry, while for the triplet anti-symmetry is required. Set the EV

$$w_{\mathbf{k}} = \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \tag{1.16}$$

Let us now break down the MFT discussion for the local and non-local interactions.

Local repulsion. The detailed treatment of $\hat{H}_U$ is discussed in Sec. ??, in the context of conventional HM. We here just sketch the most important results. By contracting in the Cooper channel the Fourier transform of $\hat{H}_U$ of Eq. (??),

$$\hat{H}_U \stackrel{\text{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}$$

in the hypothesis that preserving space translations invariance the net momentum of the Cooper pair is null, as in Eq. (??) the local part of the hamiltonian is reduced to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w_{\mathbf{k}'}^* \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (1.17)$$

being $E_{C/U}^{(\text{SC})}$ the Cooper contraction given in Eq. (??). The Hartree channel is discussed in Sec. ??.

Let now:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} w_{\mathbf{k}}$$

which reduces Eq. (1.17) to

$$\hat{H}_U \simeq (\text{Hartree channel}) + U \sum_{\mathbf{k} \in \text{BZ}} \left[w^{(s)} \hat{\phi}_{\mathbf{k}} + w^{(s)*} \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (1.18)$$

As in Eq. (??), this will produce the s -wave part of the gap function.

Non-local attraction. Consider Eq. (??), and perform Wick contractions in the Cooper channel, giving

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow} \right. \\ & \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow} \rangle \right] \end{aligned}$$

Identical considerations as above hold, and just the $\mathbf{K} = \mathbf{0}$ term contributes.:

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \right. \\ & \left. + \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \rangle - \langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \rangle \right] \end{aligned}$$

Recognizing our superconducting order parameter in the various EVs, we get

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k}, \mathbf{k}'} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w_{\mathbf{k}'}^* \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/V/\text{o.s.}}^{(\text{SC})} \quad (1.19)$$

We also made use of the harmonics decomposition of Eq. (??). In the above equation, the contraction energy is given by

$$E_{C/V/\text{o.s.}}^{(\text{SC})} \equiv - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k}, \mathbf{k}'} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} w_{\mathbf{k}} w_{\mathbf{k}'}^* \quad (1.20)$$

Let then the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} w_{\mathbf{k}}$$

Eq. (1.19) becomes

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - V \sum_{\gamma} \sum_{\mathbf{k} \in \text{BZ}} \left[w^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E_{C/V/\text{o.s.}}^{(\text{SC})}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

Symmetry	Self-consistency equation	Graph
s -wave	$w^{(s)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
s^* -wave	$w^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_x -wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_y -wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
$d_{x^2-y^2}$ -wave	$w^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??

Table 1.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$.

For the γ -wave component of the superconducting order parameter, the following chain, similar to Eq. (??), holds:

$$\begin{aligned}
w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\langle \hat{s}_{\mathbf{k}}^x \rangle + i \langle \hat{s}_{\mathbf{k}}^y \rangle}{2} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta]
\end{aligned} \tag{1.21}$$

where $\tau^+ \equiv (\tau^x + i\tau^y)/2$. We used Eqns. (1.3), (1.4) and (1.6) as well as the thermal factor defined in Eq. (1.7). The derivation above holds for all symmetries, including the s one as testifies Eq. (??). Explicitly, five self-consistency equations are then derived and listed in Tab. 1.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter just in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

1.4.2 Cooper o.s. contraction energy

The Cooper contraction energy is immediately derived from Eq. (1.20) by recognizing the harmonics of $w_{\mathbf{k}}$,

$$E_{C/V/o.s.}^{(\text{SC})} = -L_x L_y \times V \sum_{\gamma} |w^{(\gamma)}|^2$$

and is completed by its local counterpart

$$E_{C/U}^{(\text{SC})} = L_x L_y \times U |w^{(s)}|^2$$

discussed for the conventional HM in Eq. (??).

1.4.3 Cooper s.s. channel in the SC phase: analytical considerations

Consider Eq. (??) in the Cooper channel,

$$\begin{aligned} \hat{H}_V^{(s.s.)} \simeq & (\text{Hartree channel}) + (\text{Fock channel}) \\ & - \frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \right. \\ & \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \right] \end{aligned}$$

In principle, this pairing channel is legitimate and gives rise to “polarized” (same spin) Cooper class RWCs. This kind of spin pairing couples with antisymmetric spatial structures, working in the triplet pairing channel. However, we can safely ignore and set to zero these EVs, as is proven [14], [10]. The reason is simply that the superconducting phase preserves SU(2) spin symmetry, thus all three triplet pairing subspaces

$$z \text{ spin projection} \in \{-1, 0, 1\}$$

are degenerate and give rise to an identical solution. We can safely use the most convenient to us, the one with null spin projection, which can be mapped on the HM model of superconductivity discussed in Sec. ??.

Anyhow, for the sake of completeness, the s.s. pairing channel is explicitly discussed in Sec. [Missing]. As is seen there, following the sketch of [10], the resulting triplet of HFPs can be safely reduced to the simplified set of o.s. triplet HFPs discussed in previous sections.

1.5 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_c^{(\text{SC})} \equiv - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} + E_{\text{H}/V}^{(\text{SC})} + E_{\text{F}/V}^{(\text{SC})} + E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} \right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{-\mathbf{k}\downarrow} \right] - \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\Delta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \tilde{\Delta}_{\mathbf{k}}^* \hat{\phi}_{\mathbf{k}}^{\dagger} \right] + E_a^{(\text{SC})} + E_c^{(\text{SC})}$$

having we included the “anticommutation energy” also defined for the conventional HM in Eq. (??),

$$E_a^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V \sum_{\gamma \in \{s^*, d\}} u^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} - \tilde{\mu} \quad (1.22)$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -U w^{(s)} + V \sum_{\gamma} w^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \quad (1.23)$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -U w^{(s)} \quad \tilde{\Delta}^{(\gamma)} = V w^{(\gamma)} \quad \text{being } \gamma \in \{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)} (\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)} (\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2} \left[\tilde{\Delta}^{(p_x)} \sin k_x + \tilde{\Delta}^{(p_y)} \sin k_y \right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalized ground-state.

1.5.1 The relative phase in the zero temperature half-filled ground state

The conventional BCS ground state is widely known and derived for HM superconductivity in Sec. ???. Jumping to Eq. (??) and extending it to the present context of the EHM, we get

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} \left[\cos \tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}} \sin \tilde{\theta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^\dagger \right] |\Omega_{\mathbf{k}}\rangle$$

a BCS state described by the RMPs. Within this short section, we just illustrate the most interesting features of this state. First, the phase factor angle is the gap “tilt” in the complex plane. This is evident by the facts

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}}^x | \Psi \rangle &= \langle \Psi | \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^\dagger | \Psi \rangle \\ &= \sin \tilde{\theta}_{\mathbf{k}} \cos \tilde{\theta}_{\mathbf{k}} \left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}} \right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \cos(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}}^y | \Psi \rangle &= i \langle \Psi | \hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^\dagger | \Psi \rangle \\ &= i \sin \tilde{\theta}_{\mathbf{k}} \cos \tilde{\theta}_{\mathbf{k}} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}} \right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \sin(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

results that are coherent with Eqns. (1.3) as well as (1.4) and allow us to identify the phase with the pseudofield xy angle.

1.6 SC free energy density

Free energy density of the superconducting phase is derived as for the normal phase in Sec. ???. As for the AF phase of Sec. ??, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (??), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right) \ln \left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right) + f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \ln f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right] \quad (1.24)$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^x + 1} + \frac{1}{e^{-x} + 1} = 1$$

it's immediate to conclude that

$$s^{(-)} = s^{(+)} \implies s \equiv s^{(-)} + s^{(+)} = 2s^{(+)}$$

being s the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:

$$\begin{aligned} e_g^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} \tilde{\text{T}}\text{h}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

2. Anticommutation energy:

$$\begin{aligned} e_a^{(\text{SC})} &\equiv \frac{E_a^{(\text{SC})}}{L_x L_y} \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}} \end{aligned}$$

3. Contraction energy:

$$e_c^{(\text{SC})} = -U \left[n^2 + |w^{(s)}|^2 \right] + V \left[2zn^2 - \sum_{\gamma} |u^{(\gamma)}|^2 + \sum_{\gamma} |w^{(\gamma)}|^2 \right]$$

4. Entropic energy

$$e_s^{(\text{SC})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}}; \beta, 0)$$

5. Chemical potential correction

$$e_n^{(\text{SC})} = \mu n$$

Compose everything

$$f_{\text{MFT}}^{(\text{SC})} \equiv e_g^{(\text{SC})} + e_a^{(\text{SC})} + e_c^{(\text{SC})} + e_s^{(\text{SC})} + e_n^{(\text{SC})}$$

and use that:

$$\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x + 1} = 1$$

In the end we get:

$$\begin{aligned} f_{\text{MFT}}^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}} \right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \\ &\quad - U \left[n^2 + |w^{(s)}|^2 \right] + V \left[2zn^2 - \sum_{\gamma} |u^{(\gamma)}|^2 + \sum_{\gamma} |w^{(\gamma)}|^2 \right] + \mu \end{aligned}$$

As was for the normal and AF phases, we recognize the chemical potential shift of Eq. (1.9),

$$\begin{aligned} f_{\text{MFT}}^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}} \right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \\ &\quad - U |w^{(s)}|^2 - V \left[\sum_{\gamma} |u^{(\gamma)}|^2 - \sum_{\gamma} |w^{(\gamma)}|^2 \right] + \tilde{\mu} n \quad (1.25) \end{aligned}$$

1.6.1 The gap energy

Similarly to Sec. ??, let us prove the following equation about the gap energy:

$$-U |w^{(s)}|^2 + V \sum_{\gamma} |w^{(\gamma)}|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \quad (1.26)$$

Expand the right-hand side through Eq. (1.23),

$$\begin{aligned}
\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[-U w^{(s)*} + V \sum_{\gamma} w^{(\gamma)*} \varphi_{\mathbf{k}}^{(\gamma)} \right] \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \\
&= -U w^{(s)*} \times \overbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}} \\
&\quad + V \sum_{\gamma} w^{(\gamma)*} \times \underbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}
\end{aligned}$$

having we used the self-consistency equations of Tab. 1.5. This proves Eq.(1.26), which is pretty similar to Eq. (??) with one important exception. Since the right-hand side of Eq.(1.26) is positive by definition, for $V > 0$ we have the constraint

$$\sum_{\gamma} |w^{(\gamma)}|^2 \geq M \quad \text{being} \quad M^{(\text{SC})} \equiv |w^{(s)}|^2 \frac{U}{V}$$

Now, as we state in Sec. ??, for the pure HM we have identically $w^{(s)} = 0$. The pure HM cannot express BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U \ll V$, even small s -wave gaps impose larger $\gamma \neq s$ components;
- The s -wave gap $w^{(s)}$ cannot rise without the contributions of more complex gap topologies, which translates in (with an uncivil abuse of notation)

$$\bigotimes_{\gamma \neq s} \lim_{w^{(\gamma)} \rightarrow 0} w^{(s)} = 0$$

The unconventional limit notation indicates a limit being taken contemporarily on all symmetries except the s -wave one.

Of course, Eq. (1.26) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

1.7 Numerical simulations

[To be continued...]

1.7.1 HF iterative algorithm sketch

[To be continued...]

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